VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD
B.TECH. MINOR IN ARITFICIAL INTELLIGENCE AND MACHINE LEARNING

COURSE STRUCTURE AND SYLLABUS
(Applicable for the batches admitted from the academic year 2022-2023)

V SEMESTER **R22**

Course Code	Title of the Course	L	T	P/D	CH	C
22MC1AM301	Foundations of Artificial Intelligence	3	0	0	3	3
22MC2AM301	Artificial Intelligence Laboratory	0	0	3	3	1.5
Total		**3**	**0**	**3**	**6**	**4.5**

VI SEMESTER **R22**

Course Code	Title of the Course	L	T	P/D	CH	C
22MC1AM302	Artificial Intelligence Applications	3	1	0	4	4
Total		**3**	**1**	**0**	**4**	**4**

VII SEMESTER **R22**

Course Code	Title of the Course	L	T	P/D	CH	C
22MC1AM401	Principles of Machine Learning	3	0	0	3	3
22MC1AM402	Fundamentals of Deep Learning					
22MC2AM401	Principles of Machine Learning Laboratory	0	0	3	3	1.5
22MC2AM402	Fundamentals of Deep Learning Laboratory					
Total		**3**	**0**	**3**	**6**	**4.5**

VII SEMESTER **R22**

Course Code	Title of the Course	L	T	P/D	CH	C
22MC1AM403	Robotics Process Automation	3	0	0	3	3
22MC1AM404	Natural Language Processing					
22MC1AM405	Game Theory					
22MC1AM406	Computer Vision and Robotics					
22MC1AM407	Speech and Video Processing					
22MC1AM408	Soft Computing					
22MP1AM401	Mini – Project	0	0	4	4	2
Total		**3**	**0**	**4**	**7**	**5**

L – Lecture T – Tutorial P – Practical D – Drawing CH – Contact Hours/Week
C – Credits SE – Sessional Examination CA – Class Assessment ELA – Experiential Learning Assessment
SEE – Semester End Examination D-D – Day to Day Evaluation LR – Lab Record
CP – Course Project PE – Practical Examination

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (AIML) V Semester

(22MC1AM301) FOUNDATIONS OF ARTIFICIAL INTELLIGENCE

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:
- To review important concepts required for Artificial Intelligence
- To introduce the concept of learning patterns from data and develop a strong model using machine learning algorithms
- To explain theoretical foundation for understanding state of the art Machine Learning algorithms like classifications, regressions
- To understand unsupervised learning technique using partitioning and hierarchical clustering methods
- To gain the knowledge on clustering techniques

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Represent knowledge and facts for solving the real-world problems
CO-2: Build, and Evaluate the model using different Machine Learning techniques
CO-3: Design and implement machine learning solutions to classification, regression
CO-4: Design and implement various unsupervised learning methods to real-world applications
CO-5: Implement clustering techniques on real time data

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12
CO-1	3	1	2	2	3	1	2	-	1	-	-	1
CO-2	2	3	2	3	2	1	2	-	2	-	-	2
CO-3	2	2	3	3	3	2	2	-	2	-	-	3
CO-4	2	2	3	3	2	1	2	-	2	1	-	2
CO5	3	2	3	2	3	2	2	1	3	3	3	3

UNIT-I:
Defining Artificial Intelligence, Defining AI techniques, Using Predicate Logic and Representing Knowledge as Rules, Representing simple facts in logic, Computable functions and predicates, Procedural vs Declarative knowledge, Logic Programming.

UNIT-II:
Mathematical Foundations: Matrix Theory and Statistics for Machine Learning. Idea of Machines learning from data, Classification of problem – Regression and Classification, Supervised and Unsupervised learning.

UNIT-III:
Linear Regression: Model representation for single variable, Single variable Cost Function, Gradient Decent for Linear Regression, Gradient Decent in practice

UNIT-IV:
Logistic Regression: Classification, Hypothesis Representation, Decision Boundary, Cost function, Advanced Optimization, Multi-classification (One vs All), Problem of Overfitting.

UNIT-V:
Clustering: clustering as a machine learning task, different types of clustering techniques, partitioning methods, k-medoids, hierarchical clustering. Use-cases centered around classification and clustering.

TEXT BOOKS:
1. Artificial Intelligence, Cengage Learning, Saroj Kaushik, 1st Edition, 2011
2. Python Machine Learning by Example, Yuxi (Hayden) Liu, Packet Publishing, 2017

REFERENCES:
1. Machine Learning, Saikar Dutt, Subramanian Chandramouli, Amit Kumar Das, Pearson India
2. Practical Workbook Artificial Intelligence and Soft Computing for Beginners, Anindita Das Bhattacharjee, Shroff Publisher-X team Publisher
3. Machine Learning, Tom Mitchell, McGraw-Hill, 2017
4. Pattern Recognition and Machine Learning, Christopher M. Bishop, Springer, 2011
5. The Elements of Statistical Learning, T. Hastie, R. Tibshirani, J. Friedman, 2nd Edition, 2011

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (AIML) V Semester

(22MC2AM301) ARTIFICIAL INTELLIGENCE LABORATORY

TEACHING SCHEME		
L	T/P	C
0	3	1.5

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To review important concepts required for Artificial Intelligence
- To introduce the concept of learning patterns from data and develop a strong model using machine learning algorithms
- To explain theoretical foundation for understanding state of the art Machine Learning algorithms like classifications, regressions
- To understand unsupervised learning technique using partitioning and hierarchical clustering methods
- To explore various statistical methods for machine learning

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Represent knowledge and facts for solving the real-world problems
CO-2: Build, and evaluate the model using different machine learning techniques
CO-3: Design and implement machine learning solutions to classification, regression
CO-4: Design and implement various unsupervised learning methods to real-world applications
CO-5: Apply statistical methods to develop the machine learning models

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12
CO-1	3	1	2	2	2	1	2	-	1	-	-	1
CO-2	2	3	3	3	3	1	2	-	2	-	-	2
CO-3	3	2	3	3	2	2	2	-	2	-	-	2
CO-4	3	3	3	3	3	1	2	-	2	1	3	2
CO-5	3	2	2	3	3	2	2	1	3	3	3	2

LIST OF EXPERIMENTS:

WEEK-1 & 2: Basic programs in Python to get familiarize various programming structures

WEEK - 3: Implementation of logical rules in Python

WEEK - 4, 5, 6 & 7: Using any data apply the concept of:
a. Liner regression
b. Gradient decent
c. Logistic regression

WEEK - 8: Perform and plot overfitting in a data set

WEEK - 9 &10: Implementation of KNN classification algorithm

WEEK - 11 &12: Implementation of k-means clustering algorithm

WEEK-13: Explore statistical methods for machine learning

TEXT BOOKS:

1. Artificial Intelligence, Cengage Learning, Saroj Kaushik, 1st Edition, 2011
2. Python Machine Learning by Example, Yuxi (Hayden) Liu, Packet Publishing, 2017
3. Machine Learning, Saikar Dutt, Subramanian Chandramouli, Amit Kumar Das, Pearson India

REFERENCES:

1. Practice Workbook: Artificial Intelligence and Soft Computing for Beginners, Anindita Das Bhattacharjee, 1st Edition, Shroff/X-Team, 2018
2. Machine Learning, Tom Mitchell, McGraw-Hill, 2017
3. Pattern Recognition and Machine Learning, Christopher M. Bishop, Springer, 2011
4. The Elements of Statistical Learning, T. Hastie, R. Tibshirani, J. Friedman, 2nd Edition, 2011

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (AIML) VI Semester

(22MC1AM302) ARTIFICIAL INTELLIGENCE APPLICATIONS

TEACHING SCHEME		
L	T/P	C
3	1	4

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To give deep knowledge of AI
- To learn how AI can be applied in various fields to make the life easy
- To introduce recent topics in AIML
- To gain and knowledge about Robotic Process automation, and AI optimized hardware
- To understand the recent developments in Artificial Intelligence

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: correlate the AI and solutions to modern problem
CO-2: Decide when to use which type of AI technique
CO-3: Understand and analyse real world problems using AI techniques
CO-4: Develop solutions using Artificial Intelligence and block chain technologies
CO-5: Build smart solutions using AI techniques

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12
CO-1	1	1	3	2	2	2	2	-	1	-	-	1
CO-2	3	3	3	3	2	1	2	-	2	-	-	2
CO-3	3	2	3	3	2	2	2	-	2	-	-	2
CO-4	3	2	3	3	3	1	3	-	2	-	-	2
CO-5	3	3	3	3	2	2	2	1	3	3	3	2

UNIT-I:
Linguistic aspects of natural language processing, A.I. And Quantum Computing, Applications of Artificial Intelligence (AI) in business.

UNIT-II:
Emotion Recognition using human face and body language, AI based system to predict the diseases early,

UNIT-III:
Smart Investment analysis, AI in Sales and Customer Support, Robotic Processes Automation for supply chain management.

UNIT-IV:
AI-Optimized Hardware, Digital Twin i.e. AI Modelling, Information Technology & Security using AI.

UNIT-V:

Recent Topics in AI/ML: AI/ML in Smart solutions, AI/ML in Social Problems handling, Block chain and AI.

TEXT BOOKS:

1. AI and Analytics, Accelerating Business Decisions, Sameer Dhanrajani, John Wiley
2. Artificial Intelligence in Practice: How 50 Successful Companies Used AI and Machine Learning to Solve Problems, Bernard Marr, Matt Ward, Wiley

REFERENCES:

1. Life 3.0: Being Human in the Age of Artificial Intelligence, Max Tegmark, 2018
2. Homo Deus: A Brief History of Tomorrow, Yuval Noah Harari, 2017

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (AIML) VII Semester

(22MC1AM401) PRINCIPLES OF MACHINE LEARNING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Data Structures, Knowledge on Statistical Methods

COURSE OBJECTIVES:
- To discuss fundamental concepts of machine learning techniques
- To discuss machine learning techniques such as decision tree learning, Bayesian learning
- To understand computational learning theory
- To study the pattern comparison techniques
- To examine analytical learning, explanation-based learning, and the integration of inductive and analytical learning approaches

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand the concepts of computational intelligence like machine learning
CO-2: Get the skill to apply machine learning techniques to address the real time problems in different areas
CO-3: Understand the Neural Networks and its usage in machine learning application.
CO-4: Analyse and build the models using decision tree and Bayesian learning techniques
CO-5: Understand and apply analytical learning methods, combining inductive and analytical learning strategies effectively

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12
CO-1	3	2	1	1	2	1	1	1	2	1	1	2
CO-2	2	3	2	2	3	2	1	1	2	2	1	2
CO-3	3	3	3	2	2	1	2	1	2	3	2	2
CO-4	3	2	2	2	3	2	1	2	3	3	2	2
CO-5	3	3	3	3	3	2	2	2	3	3	3	3

UNIT-I:
Introduction - Well-posed learning problems, designing a learning system, Perspectives and issues in machine learning, Concept learning and the general to specific ordering – introduction, a concept learning task, concept learning as search, find-S: finding a maximally specific hypothesis, version spaces and the candidate elimination algorithm, remarks on version spaces and candidate elimination, inductive bias. Decision Tree Learning – Introduction, decision tree representation, appropriate problems for decision tree learning, the basic decision tree learning

algorithm, hypothesis space search in decision tree learning, inductive bias in decision tree learning, issues in decision tree learning.

UNIT-II:
Artificial Neural Networks-1– Introduction, neural network representation, appropriate problems for neural network learning, perceptions, multilayer networks and the back-propagation algorithm. Artificial Neural Networks-2- Remarks on the Back-Propagation algorithm, an illustrative example: face recognition, advanced topics in artificial neural networks. Evaluation Hypotheses – Motivation, estimation hypothesis accuracy, basics of sampling theory, a general approach for deriving confidence intervals, difference in error of two hypotheses, comparing learning algorithms.

UNIT-III:
Bayesian learning – Introduction, Bayes theorem, Bayes theorem and concept learning, Maximum Likelihood and least squared error hypotheses, maximum likelihood hypotheses for predicting probabilities, minimum description length principle, Bayes optimal classifier, Gibs algorithm, Naïve Bayes classifier, an example: learning to classify text, Bayesian belief networks, the EM algorithm.
Computational learning theory – Introduction, probably learning an approximately correct hypothesis, sample complexity for finite hypothesis space, sample complexity for infinite hypothesis spaces, the mistake bound model of learning. Instance-Based Learning- Introduction, k-nearest neighbour algorithm, locally weighted regression, radial basis functions, case-based reasoning, remarks on lazy and eager learning.

UNIT- IV:
Genetic Algorithms – Motivation, Genetic algorithms, an illustrative example, hypothesis space search, genetic programming, models of evolution and learning, parallelizing genetic algorithms. Learning Sets of Rules – Introduction, sequential covering algorithms, learning rule sets: summary, learning First-Order rules, learning sets of First-Order rules: FOIL, Induction as inverted deduction, inverting resolution. Reinforcement Learning – Introduction, the learning task, Q–learning, non-deterministic, rewards and actions, temporal difference learning, generalizing from examples, relationship to dynamic programming.

UNIT-V:
Analytical Learning-1- Introduction, learning with perfect domain theories: PROLOG-EBG, remarks on explanation-based learning, explanation-based learning of search control knowledge. Analytical Learning-2-Using prior knowledge to alter the search objective, using prior knowledge to augment search operators. Combining Inductive and Analytical Learning – Motivation, inductive-analytical approaches to learning, using prior knowledge to initialize the hypothesis.

TEXT BOOK:
1. Machine Learning, Tom M. Mitchell, 1st Edition, McGraw-Hill, 2017

REFERENCE:
1. Machine Learning: An Algorithmic Perspective, Stephen Marshland, 2nd Edition, CRC/Taylor & Francis, 2014

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (AIML) VII Semester

(22MC1AM402) FUNDAMENTALS OF DEEP LEARNING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand complexity of Deep Learning algorithms and their limitations
- To be capable of performing experiments in Deep Learning using real-world data
- To discuss convolution neural network and computer vision concepts
- To learn fundamental concepts of Deep Learning techniques
- To examine advanced NLP tasks, including named entity recognition, sentiment analysis, and dialogue generation using deep learning models like RNNs and CNNs

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Implement deep learning algorithms, understand neural networks and traverse the layers of data
CO-2: Learn concepts such as convolutional neural networks, recurrent neural networks, training deep networks and high-level interfaces
CO-3: Understand applications of Deep Learning to Computer Vision
CO-4: Understand and analyse applications of Deep Learning to NLP
CO-5: Develop expertise in solving advanced NLP problems, such as sentiment analysis, entity recognition, and dialogue generation with LSTMs

UNIT-I:
Introduction: Feed forward Neural networks, Gradient descent and the back propagation algorithm, Unit saturation, the vanishing gradient problem, and ways to mitigate it. RelU Heuristics for avoiding bad local minima, Heuristics for faster training, Nestors accelerated gradient descent, Regularization, Dropout

UNIT-II:
Convolutional Neural Networks: Architectures, convolution/pooling layers, Recurrent Neural Networks: LSTM, GRU, Encoder Decoder architectures. Deep Unsupervised Learning: Auto encoders, Variational Auto-encoders, Adversarial Generative Networks, Auto-encoder and DBM Attention and memory models, Dynamic Memory Models.

UNIT-III:
Applications of Deep Learning to Computer Vision: Image segmentation, object detection, automatic image captioning, Image generation with Generative adversarial networks, video to text with LSTM models, Attention Models for computer vision tasks

UNIT-IV:
Applications of Deep Learning to NLP: Introduction to NLP and Vector Space Model of Semantics, Word Vector Representations: Continuous Skip-Gram Model, Continuous Bag-of-Words model (CBOW), Glove, Evaluations and Applications in word similarity

UNIT-V:
Analogy Reasoning: Named Entity Recognition, Opinion Mining using Recurrent Neural Networks: Parsing and Sentiment Analysis using Recursive Neural Networks: Sentence Classification using Convolutional Neural Networks, Dialogue Generation with LSTMs

TEXT BOOKS:

1. Deep Learning, Ian Goodfellow, Yoshua Bengio and Aaron Courville, MIT Press
2. The Elements of Statistical Learning, T. Hastie, R. Tibshirani, and J. Friedman, Springer
3. Probabilistic Graphical Models, Koller and N. Friedman, MIT Press

REFERENCES:

1. Pattern Recognition and Machine Learning, Bishop C. M., Springer, 2006
2. Artificial Neural Networks, Yegnanarayana B., PHI Learning, 2009
3. Matrix Computations, Golub G. H., and Van Loan C. F., JHU Press, 2013
4. Neural Networks: A Classroom Approach, Satish Kumar, Tata McGraw-Hill Education, 2004

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (AIML) VII Semester

(22MC2AM401) PRINCIPLES OF MACHINE LEARNING LABORATORY

TEACHING SCHEME		
L	T/P	C
0	3	1.5

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:
- To get an overview of the various machine learning techniques
- To demonstrate machine learning techniques using python
- To understand how to build the model using machine learning techniques
- To discuss genetic technique
- To discuss backpropagation technique

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand complexity of Machine Learning algorithms and their limitations
CO-2: Understand modern notions in data analysis-oriented computing
CO-3: Confidently apply common Machine Learning algorithms in practice
CO-4: Perform experiments in Machine Learning using real-world data
CO-5: Implement their own approach of dealing with real-world data

LIST OF EXPERIMENTS:

1. The probability that it is Friday and that a student is absent is 3 %. Since there are 5 school days in a WEEK, the probability that it is Friday is 20 %. What is the probability that a student is absent given that today is Friday? Apply Baye's rule in python to get the result. (Ans: 15%)
2. Extract the data from database using python.
3. Implement k-nearest neighbours classification using python
4. Given the following data, which specify classifications for nine combinations of VAR1 and VAR2 predict a classification for a case where VAR1=0.906 and VAR2=0.606, using the result of k- means clustering with 3 means (i.e., 3 centroids)

VAR1	VAR2	CLASS
1.713	1.586	0
0.180	1.786	1
0.353	1.240	1
0.940	1.566	0
1.486	0.759	1
1.266	1.106	0
1.540	0.419	1
0.459	1.799	1
0.773	0.186	1

1. The following training examples map descriptions of individuals onto high, medium and low credit-worthiness.

medium skiing design single twenties no -> highRisk
high golf trading married forties yes -> lowRisk
low speedway transport married thirties yes -> medRisk

medium football banking single thirties yes -> lowRisk
high flying media married fifties yes -> highRisk
low football security single twenties no -> medRisk
medium golf media single thirties yes -> medRisk
medium golf transport married forties yes -> lowRisk
high skiing banking single thirties yes -> highRisk
low golf unemployed married forties yes -> highRisk
Input attributes are (from left to right) income, recreation, job, status, age-group, home-owner. Find the unconditional probability of `golf' and the conditional probability of `single' given `medRisk' in the dataset?

2. Implement linear regression using python.
3. Implement Naïve Bayes theorem to classify the English text
4. Implement an algorithm to demonstrate the significance of genetic algorithm
5. Implement the finite words classification system using Back-propagation algorithm

TEXT BOOK:

1. Machine Learning, Tom M. Mitchell, 1st Edition, McGraw-Hill, 2017

REFERENCE:

1. Machine Learning: An Algorithmic Perspective, Stephen Marshland, 2nd Edition, CRC/Taylor & Francis, 2014

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (AIML) VII Semester

(22MC2AM402) FUNDAMENTALS OF DEEP LEARNING LABORATORY

TEACHING SCHEME		
L	T/P	C
0	3	1.5

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:
- To build the foundation of Deep Learning
- To understand how to build the Neural Network
- To enable students to develop successful machine learning concepts
- To discuss fundamentals concepts of deep learning techniques in the area of Natural language Processing
- To build the natural language processing applications using deep learning concepts

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Learn the fundamental principles of Deep Learning
CO-2: Identify the Deep Learning algorithms for various types of learning tasks in various domains
CO-3: Implement Deep Learning algorithms and solve real-world problems
CO-4: Apply Deep Learning methods to natural language processing applications
CO-5: Train and evaluate sentiment analysis models using RNN layers with LSTM/GRU to classify text data effectively

LIST OF EXPERIMENTS:

1. Setting up the Spyder IDE Environment and Executing a Python Program
2. Installing Keras, Tensorflow and Pytorch libraries and making use of them
3. Applying the Convolution Neural Network on computer vision problems
4. Image classification on MNIST dataset (CNN model with Fully connected layer)
5. Applying the Deep Learning Models in the field of Natural Language Processing
6. Train a sentiment analysis model on IMDB dataset, use RNN layers with LSTM/GRU notes
7. Applying the Autoencoder algorithms for encoding the real-world data
8. Applying Generative Adversial Networks for image generation and unsupervised tasks.

TEXT BOOKS:

1. Deep Learning, Ian Goodfellow, Yoshua Bengio and Aaron Courville, MIT Press
2. The Elements of Statistical Learning, T. Hastie, R. Tibshirani, and J. Friedman, Springer
3. Probabilistic Graphical Models, Koller and N. Friedman, MIT Press

REFERENCES:

1. Pattern Recognition and Machine Learning, Bishop C. M., Springer, 2006
2. Artificial Neural Networks, Yegnanarayana B., PHI Learning, 2009
3. Matrix Computations, Golub G. H. and Van Loan C. F., JHU Press, 2013
4. Neural Networks: A Classroom Approach, Satish Kumar, Tata McGraw-Hill Education, 2004

EXTENSIVE READING:

1. http://www.deeplearning.net
2. https://www.deeplearningbook.org/
3. https://developers.google.com/machine-learning/crash-course/ml-intro
4. www.cs.toronto.edu/~fritz/absps/imagenet.pdf
5. http://neuralnetworksanddeeplearning.com

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (AIML) VIII Semester

(22MC1AM403) ROBOTICS PROCESS AUTOMATION

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To make learners familiar with the concepts of Robotic Process Automation
- To learn about the topics related to web control room and client
- To get knowledge on various devices and workload
- To understand the information related to bots
- To teach automation commands, error handling, and integrations for building robust bots

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Describe RPA, where it can be applied and how it's implemented
CO-2: Identify and understand Web Control Room and Client Introduction
CO-3: Understand how to handle various devices and the workload
CO-4: Understand Bot creators, Web recorders and task editors
CO-5: Manage workload, device pools, and perform administrative tasks within the platform

UNIT-I:
Introduction to Robotic Process Automation & Bot Creation Introduction to RPA and Use cases – Automation Anywhere Enterprise Platform – Advanced features and capabilities – Ways to create Bots

UNIT-II:
Web Control Room and Client Introduction - Features Panel - Dashboard (Home, Bots, Devices, Audit, Workload, Insights) - Features Panel – Activity (View Tasks in Progress and Scheduled Tasks) - Bots (View Bots Uploaded and Credentials)

UNIT-III:
Devices (View Development and Runtime Clients and Device Pools) - Workload (Queues and SLA Calculator) - Audit Log (View Activities Logged which are associated with Web CR)
Administration (Configure Settings, Users, Roles, License and Migration) -

UNIT-IV:
Demo of Exposed API's – Conclusion – Client introduction and Conclusion.
Bot Creator Introduction – Recorders – Smart Recorders – Web Recorders – Screen Recorders.

UNIT-V:
Task Editor – Variables - Command Library – Loop Command – Excel Command – Database Command - String Operation Command - XML Command.

Terminal Emulator Command - PDF Integration Command - FTP Command - PGP Command - Object Cloning Command - Error Handling Command - Manage Windows Control Command - Workflow Designer - Report Designer.

TEXT BOOK:

1. Learning Robotic Process Automation: Create Software Robots and Automate Business Processes with the Leading RPA Tool – Uipath, Alok Mani Tripathi, Packt Publishing, 2018

REFERENCE:

1. Robotic Process Automation a Complete Guide, Gerardus Blokdyk, Kindle Edition, 5StarCooks, 2020

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (AIML) VIII Semester

(22MC1AM404) NATURAL LANGUAGE PROCESSING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To explain text normalization techniques and n-gram language model
- To discuss part of speech methods and naïve bayes classification techniques
- To understand word sense disambiguation techniques and process of building question answering system
- To introduce the concepts of chatbots, dialogue systems, speech recognition systems and text to speech recognition methods
- To develop chatbots and dialogue systems, including speech recognition and synthesis integration

COURSE OUTCOMES: After the completion of the course, the student should be able to

CO-1: Apply normalization techniques on a document and evaluate a language model

CO-2: Implement parts of speech tagging and classification techniques on the words

CO-3: Establish relationships among words of a sentence using word net and also build the question answering system

CO-4: Analyze chatbots, dialogue systems, and automatic speech recognition systems

CO-5: Design and develop chatbots, dialogue systems, and integrate automatic speech recognition and text-to-speech technologies

UNIT-I:

Introduction, Regular Expressions, Text Normalization, Edit Distance: Words, Corpora, Text Normalization, Word Normalization, Lemmatization and Stemming, Sentence Segmentation, The Minimum Edit Distance Algorithm.

UNIT-II:

N-gram Language Models: N-Grams, Evaluating Language Model, Sampling sentences from a language model

Sequence Labeling for Parts of Speech and Named Entities: Part-of-Speech Tagging, Named Entities and Named Entity Tagging

UNIT-III:

Naive Bayes and Sentiment Classification: Naive Bayes Classifiers, Training the Naive Bayes Classifier, Optimizing for Sentiment Analysis, Naive Bayes as a Language Model, Evaluation: Precision, Recall, F-measure, Test sets and Cross-validation

Word Senses and WordNet: Word Senses, Relations between Senses, WordNet: A Database of Lexical Relations, Word Sense Disambiguation, WSD Algorithm: Contextual Embeddings

UNIT-IV:
Question Answering: Information Retrieval, IR-based Factoid Question Answering, IR-based QA: Datasets, Entity Linking, Knowledge-based Question Answering, Using Language Models to do QA, Classic QA Models

UNIT-V:
Chatbots & Dialogue Systems: Properties of Human Conversation, Chatbots, GUS: Simple Frame-based Dialogue Systems, The Dialogue-State Architecture, Evaluating Dialogue Systems, Dialogue System Design,
Automatic Speech Recognition and Text-to-Speech: The Automatic Speech Recognition Task, Feature Extraction for ASR: Log Mel Spectrum, Speech Recognition Architecture

TEXT BOOKS:
1. Speech and Language Processing, Dan Jurafsky and James H. Martin, 3rd Edition, Pearson Publications
2. Natural Language Processing with Python, Steven Bird, Ewan Klein, and Edward Loper, O'Reilly, 2007

REFERENCES:
1. Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana
2. Foundations of Statistical Natural Language Processing, Christopher Manning and Hinrich Schütze
3. Natural Language Processing in Action- Understanding, Analysing, and Generating Text with Python, Hobson Lane, Cole Howard, Hannes Max Hapke
4. The Handbook of Computational Linguistics and Natural Language Processing (Blackwell Handbooks in Linguistics), 1st Edition

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (AIML) VIII Semester

(22MC1AM405) GAME THEORY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To explain the concepts related to game theory
- To learn knowledge and solution concepts related to games
- To get knowledge on types of game theory and solutions
- To understand the extensive games with perfect information
- To explore advanced topics such as Nash Folk theorems, trigger strategies, and the structure of subgame perfect equilibria

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand the basic concepts of game theory and solutions
CO-2: Understand different types of equilibrium interpretations
CO-3: Understand and analyze knowledge and solution concepts
CO-4: Analyze extensive games with perfect information
CO-5: Delve into advanced topics like bargaining games, repeated games, and the structure of equilibria in discounting criteria

UNIT-I:
Introduction- Game Theory, Games and Solutions Game Theory and the Theory of Competitive Equilibrium, Rational Behavior, The Steady State and Deductive Interpretations, Bounded Rationality Terminology and Notation.
Nash Equilibrium- Strategic Games, Nash Equilibrium Examples, Existence of a Nash Equilibrium, Strictly Competitive Games, Bayesian Games: Strategic Games with Imperfect Information.

UNIT-II:
Mixed, Correlated, and Evolutionary Equilibrium - Mixed Strategy Nash Equilibrium, Interpretations of Mixed Strategy Nash Equilibrium, Correlated Equilibrium, Evolutionary Equilibrium, Rationalizability and Iterated Elimination of Dominated Actions - Rationalizability Iterated Elimination of Strictly Dominated Actions, Iterated Elimination of Weakly Dominated Actions.

UNIT-III:
Knowledge and Equilibrium - A Model of Knowledge Common Knowledge, Can People Agree to Disagree? Knowledge and Solution Concepts, The Electronic Mail Game.
Extensive Games with Perfect Information - Extensive Games with Perfect Information Subgame Perfect Equilibrium, Two Extensions of the Definition of a Game, The Interpretation of a Strategy, Two Notable Finite Horizon Games

UNIT-IV:
Iterated Elimination of Weakly Dominated Strategies Bargaining Games - Bargaining and Game Theory, A Bargaining Game of Alternating Offers Subgame Perfect Equilibrium Variations and Extensions.
Repeated Games - The Basic Idea, Infinitely Repeated Games vs. Finitely Repeated Games. Infinitely Repeated Games: Definitions, Strategies as Machines,

UNIT-V:
Trigger Strategies: Nash Folk Theorems Punishing for a Limited Length of Time: A Perfect Folk Theorem for the Limit of Means Criterion Punishing the Punisher: A Perfect Folk Theorem for the Overtaking Criterion Rewarding Players Who Punish: A Perfect Folk Theorem for the Discounting Criterion The Structure of Subgame Perfect Equilibria Under the Discounting Criterion Finitely Repeated Game

TEXT BOOKS:
1. A Course in Game Theory, M. J. Osborne and A. Rubinstein, MIT Press
2. Game Theory, Roger Myerson, Harvard University Press
3. Game Theory, D. Fudenberg and J. Tirole, MIT Press

REFERENCES:
1. Theory of Games and Economic Behavior, J. von Neumann and O. Morgenstern, John Wiley and Sons
2. Games and Decisions, R. D. Luce and H. Raiffa, John Wiley and Sons
3. Game Theory, G. Owen, 2nd Edition, Academic Press

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (AIML) VIII Semester

(22MC1AM406) COMPUTER VISION AND ROBOTICS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand the fundamental concepts related to computer vision
- To get knowledge on boundary tracking techniques
- To learn Hough transform for lines, circle, and ellipse detections
- To understand the geometry of multiple views
- Implement geometric camera models and calibration techniques for real-world computer vision applications

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Implement fundamental image processing techniques required for computer vision
CO-2: Implement boundary tracking techniques
CO-3: Apply chain codes and other region descriptors, Hough Transform for line, circle, and ellipse detections
CO-4: Apply 3D vision techniques and Implement motion related techniques. And develop applications using computer vision techniques
CO-5: To delve into geometric camera models, calibration, and model-based vision for practical applications like medical imaging and robot localization

UNIT-I:
CAMERAS: Pinhole Cameras
Radiometry – Measuring Light: Light in Space, Light Surfaces, Important Special Cases. Sources, Shadows, And Shading: Qualitative Radiometry, Sources and Their Effects, Local Shading Models, Application: Photometric Stereo, Interreflections: Global Shading Models.
Color: The Physics of Color, Human Color Perception, Representing Color, A Model for Image Color, Surface Color from Image Color.

UNIT-II:
Linear Filters: Linear Filters and Convolution, Shift Invariant Linear Systems, Spatial Frequency and Fourier Transforms, Sampling and Aliasing, Filters as Templates.
Edge Detection: Noise, Estimating Derivatives, Detecting Edges.
Texture: Representing Texture, Analysis (and Synthesis) Using Oriented Pyramids, **Application:** Synthesis by Sampling Local Models, Shape from Texture.

UNIT-III:
The Geometry of Multiple Views: Two Views
Stereopsis: Reconstruction, Human Stereopsis, Binocular Fusion, Using More Cameras. Segmentation by Clustering: What Is Segmentation? Human Vision: Grouping and Getstalt,
Applications: Shot Boundary Detection and Background Subtraction, Image Segmentation by Clustering Pixels, Segmentation by Graph-Theoretic Clustering,

UNIT-IV:

Segmentation by Fitting a Model: The Hough Transform, Fitting Lines, Fitting Curves, Fitting as a Probabilistic Inference Problem, Robustness.

Segmentation and Fitting Using Probabilistic Methods: Missing Data Problems, Fitting, and Segmentation, The EM Algorithm in Practice.

Tracking With Linear Dynamic Models: Tracking as an Abstract Inference Problem, Linear Dynamic Models, Kalman Filtering, Data Association, Applications and Examples.

UNIT-V:

Geometric Camera Models: Elements of Analytical Euclidean Geometry, Camera Parameters and the Perspective Projection, Affine Cameras and Affine Projection Equations

Geometric Camera Calibration: Least-Squares Parameter Estimation, A Linear Approach to Camera Calibration, Taking Radial Distortion into Account, Analytical **Photogrammetry, An Application:** Mobile Robot Localization

Model-Based Vision: Initial Assumptions, Obtaining Hypotheses by Pose Consistency, Obtaining

Hypotheses by pose Clustering, Obtaining Hypotheses Using Invariants, Verification, and Application: Registration in Medical Imaging Systems, Curved Surfaces and Alignment.

TEXT BOOK:

1. Computer Vision – A Modern Approach, David A. Forsyth and Jean Ponce, Pearson Education, 2015

REFERENCES:

1. Computer and Machine Vision – Theory, Algorithms and Practicalities, E. R. Davies, 4th Edition, Elsevier (Academic Press), 2013
2. Digital Image Processing, R. C. Gonzalez and R. E. Woods, Addison Wesley, 2008
3. Computer Vision: Algorithms and Applications, Richard Szeliski, Springer-Verlag, 2011

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (AIML) VIII Semester

(22MC1AM407) SPEECH AND VIDEO PROCESSING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:
- To understand the mechanisms of human speech production system
- To learn the concepts related to speech analysis and speech recognition algorithms
- To get knowledge on digital video processing
- To familiarize with object tracking and video sequence concepts
- To focus on object tracking, video segmentation methods, and compression techniques for efficient video processing

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Describe the mechanisms of human speech production systems and methods for speech feature extraction
CO-2: Understand basic algorithms of speech analysis and speech recognition
CO-3: Explain basic techniques in digital video processing, including imaging characteristics and sensors
CO-4: Apply motion estimation and object tracking algorithms on video sequence
CO-5: Develop skills in object tracking, video segmentation, and interframe compression for real-time video applications

UNIT-I:
Speech processing concepts: The speech production mechanism, Discrete time speech signals, Pole-Zero modeling of speech, relevant properties of the fast Fourier transform for speech recognition, convolution, linear and nonlinear filter banks, spectral estimation of speech using DFT. Linear Prediction analysis of speech.

UNIT-II:
Speech recognition: Real and Complex Cepstrum, application of cepstral analysis to speech signal, feature extraction for speech, static and dynamic feature for speech recognition, robustness issues, discrimination in the feature space, feature selection, MFCC, LPCC, Distance measures, vector quantization models. Gaussian Mixture model, HMM.

UNIT-III:
Basics of Video Processing: Video formation, perception and representation: Principle of color video, video cameras, video display, pinhole model, CAHV model, Camera motion, Shape model, motion model, Scene model, two-dimensional motion models. Three-Dimensional Rigid Motion, Approximation of projective mapping.

UNIT-IV:
Motion Estimation Techniques: Optical flow, motion representation, motion estimation criteria, optimization methods, pixel-based motion estimation.

Block matching algorithm, gradient Based, Intensity matching, feature matching, frequency domain motion estimation, Depth from motion.
Motion analysis applications: Video Summarization, video surveillance.

UNIT-V:
Object Tracking and Segmentation: 2D and 3D video tracking, blob tracking, kernel based counter tracking, feature matching, filtering Mosaicing, video segmentation, mean shift based, active shape model, video short boundary detection. Interframe compression, Motion compensation.

TEXT BOOKS:
1. Fundamentals of Speech Recognition, L. Rabiner and B. Juang, Prentice Hall
2. Digital Video Processing, A. Murat Tekalp, Prentice Hall
3. Discrete-Time Speech Signal Processing: Principles and Practice, Thomas F. Quatieri, Coth

REFERENCES:
1. Video Processing and Communications, Yao Wang, J. Osternann and Qin Zhang, Pearson Education
2. Speech and Audio Signal Processing, B. Gold and N. Morgan, Wiley
3. Digital Image Sequence Processing, Compression, and Analysis, Todd R. Reed, CRC Press
4. Handbook of Image and Video Processing, Al Bovik, 2nd Edition, Academic Press

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (AIML) VIII Semester

(22MC1AM408) SOFT COMPUTING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:
- To familiarize with soft computing concepts
- To introduce and use the idea of fuzzy logic and use of heuristics based on human experience
- To familiarize the Neuro-Fuzzy modeling using Classification and Clustering techniques
- To learn the concepts of Genetic algorithm and its applications and acquire the knowledge of Rough Sets
- To study rough sets, rule induction, and the integration of soft computing methods for improved problem-solving

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Identify the difference between Conventional Artificial Intelligence to Computational Intelligence
CO-2: Understand fuzzy logic and reasoning to handle and solve engineering problems
CO-3: Apply the Classification and clustering techniques on various applications, and understand the advanced neural networks and its applications
CO-4: Perform various operations of genetic algorithms, Rough Sets, and comprehend various techniques to build model for various applications
CO-5: Explore rough sets, rule induction, and their integration with other soft computing techniques

UNIT-I:
Introduction to Soft Computing: Evolutionary Computing, "Soft" computing versus "Hard" computing, Soft Computing Methods,

UNIT-II:
Recent Trends in Soft Computing, Characteristics of Soft computing, Applications of Soft Computing Techniques.

UNIT-III:
Fuzzy Systems: Fuzzy Sets, Fuzzy Relations, Fuzzy Logic, Fuzzy Rule-Based Systems. Fuzzy Decision Making, Particle Swarm Optimization

UNIT-IV:
Genetic Algorithms: Basic Concepts, Basic Operators for Genetic Algorithms, Crossover and Mutation Properties, Genetic Algorithm Cycle, Fitness Function, Applications of Genetic Algorithm.

UNIT-V:

Rough Sets, Rough Sets, Rule Induction, and Discernibility Matrix, Integration of Soft Computing Techniques.

TEXT BOOKS:

1. Soft Computing – Advances and Applications, B. K. Tripathy and J. Anuradha, Cengage Learning, 2015
2. Principles of Soft Computing, 2nd Edition, S. N. Sivanandam & S. N. Deepa, Wiley India, 2008

REFERENCES:

1. Genetic Algorithms-In Search, Optimization and Machine Learning, David E. Goldberg, Pearson Education
2. Neuro-Fuzzy and Soft Computing, J. S. R. Jang, C.T. Sun and E. Mizutani, Pearson Education, 2004
3. Fuzzy Sets & Fuzzy Logic, G. J. Klir & B. Yuan, PHI, 1995
4. An Introduction to Genetic Algorithm, Melanie Mitchell, PHI, 1998
5. Fuzzy Logic with Engineering Applications, Timothy J. Ross, McGraw- Hill International Edition, 1995